8. Implement a simple stochastic part-of-speech tagging algorithm using a basic probabilistic model to assign POS tags using python.

```python
import random

# Probabilistic dictionary
pos_dict = {
    "I": ["PRP"],
    "eat": ["VB", "NN"],
    "rice": ["NN"],
    "quickly": ["RB"]
}

sentence = "I eat rice quickly"

words = sentence.split()

print("Stochastic POS Tags:")

for word in words:
    if word in pos_dict:
        tag = random.choice(pos_dict[word])
    else:
        tag = "NN"
    print(word, "->", tag)
```

```
Stochastic POS Tags:
I -> PRP
eat -> NN
rice -> NN
quickly -> RB
```